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INTERVIEW

Top 100 Amazon Data Scientist Interview Questions!

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Aspiring to be a Data Scientist at Amazon? Refer- Amazon Data Scientist Interview Questions here.

Amazon(US) tops the LinkedIn list of the top companies for 2nd time in a row! Moreover, in 2021, Amazon created more than 200,000 jobs in the US, where more than 1.1 million people are currently employed.

To add to it, a 2021 Insider survey also suggested that majority of the Amazon Employees are happy with their jobs. If you, as the many, aspire to be one of those happy Amazon employees, then get set and start preparing for the upcoming Amazon interviews!



When you start preparing for Amazon interviews, check this little guide below:

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What interviews to expect in Amazon for a Data Scientist position?

Well, the complete process of the Amazon Data Scientist interview can take anywhere between **four to eight weeks!** And what rounds to look for:

- **Emails & Applications**
- **Recruiter Interview (mostly, video call)(~30 min)**
- **Technical Round (1-2 video calls with the panel, ~60 min each)**
- **Onsite interviews (5-6 interviews, 45-60 min each)**

Have you heard about [Amazon's 16 leadership principles?](#) Make sure you align your CV with it!



Then what do you think Amazon is looking for?

1. **Technical competencies & Coding**
2. **Leadership principles**
3. **Excellent Interpersonal skills like Problem-solving, Communication, Logical Thinking, etc.**
4. **Data analysis and manipulation**
5. **Machine learning / AI**
6. **Business acumen**

And that's why you most likely will get questions in the below ratio (approximately):

1. **Coding (37%)**
2. **Machine Learning (27%)**
3. **Behavioural (19%)**
4. **Statistics (17%)**

If you are a beginner, then Amazon Data Scientist Interview Questions might include questions inclusive of (not limited to) the following:

1] Amazon Data Scientist Interview Questions for Beginners:

1. What is Data Science?
2. Tell us something about AWS.
3. Was there any time when you had to deal with ambiguity?
4. What is a confusion matrix?
5. Differentiate between supervised and unsupervised learning.



6. What are the types of selection bias?
7. Describe Markov Chains.
8. How do you find the RMSE and MSE linear regression models?
9. Why do we use a summary function?
10. Why is R used in data visualization?
11. Have you ever worked in a team?
12. Explain any of your Data Science projects in detail.
13. What are your time management strategies?
14. Name three disadvantages of using a linear model
15. Why do you need to perform resampling?

Here are some **Amazon data scientist interview questions** for the next level of interviews.

But before you rush to find them out, check here to learn more about Data Science: [What Skills Are Needed To Be A Data Scientist?](#)

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2] Amazon data scientist interview questions on Coding SQL



1. What is the most advanced query written by you?

2. Difference between CHAR and VARCHAR in MySQL
3. How do you differentiate between Oracle and MySQL?
4. How to test for NULL values in a database?
5. Write a SQL code to explain month to month user retention rate.
6. What is a foreign key? How to implement the same in MySQL?
7. What are Database White Box Testing & Black Box Testing?
8. Describe different JOINS in SQL.
9. Given a table with three columns (id, category, value) and each id has three or fewer categories (price, size, colour). How can you find those IDs for which the value of two or more categories match one another?
10. Given a CSV file with ID and Quantity columns, 50 million records, and the size of the data is 2gig, write a program to aggregate the QUANTITY column.
11. Which drivers are necessary for MySQL?
12. What is a trigger in MySQL? How can we use it?
13. How can we run MySQL in batch mode?
14. What are the various types of tables present in Mysql?
15. How can you convert timestamps to date time in MySQL?
16. How to import and export MySQL databases?

Learn more about SQL: [Top SQL Interview Questions One Needs On Their Fingertips!!!](#)

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3] Amazon Data Scientist interview questions concerning Data structure and algorithms

1. What is the computational complexity of a Python code that will help you recognize whether the entries to a list have the same characters or not
2. Suppose you have an array of integers of which you want to find a particular element; what effective algorithm would you use, and what is its efficiency?
3. For a long sorted list and a short (4 element) sorted list, what algorithm would you use to search the long list for the 4 elements?
4. Given an unfair coin with the probability of heads not equal to .5, what algorithm could you use to create a list of random 1s and 0s?
5. Given a bar plot, imagine you are pouring water from the top. How do you quantify how much water can be kept in the bar chart? ([solution](#))
6. Write a Python function that displays the first n Fibonacci numbers. ([solution](#))
7. Suppose you have a list of strings, each of which is an English sentence. # Output a dictionary out_dict that maps a key n to the list of words that occur in n different sentences. # E.g. # Input: str_list = ["The cat ate the fish", "The cat saw the roses", "The roses are red"]
8. If given an integer n and an array of numbers, state the histogram divided into n bins.

Master Data Structure & Algorithm: [Why Do You Need To Focus On Data Structure & Algorithms?](#)

4] Amazon Data Scientist interview questions regarding Modelling



1. How would you improve a classification model that suffers from low precision?
2. We have two models, one with 85% accuracy, and one with 82%. Which one do you pick? ([solution](#))
3. When you have time series data by month, and it has large data records, how will you find significant differences between this month and the previous month?
4. How do you inspect missing data, and when are they important?
5. Assume you have a file containing data in the form of `data = [{"one":a1, "two":b1,...}, {"one":a2, "two":b2,...}, {"one":a3, "two":b3,...},...]` How could you split this data into 30% test and 70% train data?

5] Amazon Data Scientist interview questions: Machine Learning

1. What is Machine Learning?
2. What is Pruning?
3. What is linear regression?
4. How do you interpret logistic regression?
5. How does dropout work?
6. What is L1 vs L2 regularization?
7. Explain the drawbacks of a linear model
8. What is the difference between bagging and boosting?
9. Explain in detail how a 1D CNN works.
10. Describe a case where you have solved an ambiguous business problem using machine learning.
11. Having a categorical variable with thousands of distinct values, how would you encode it?



12. Briefly describe the Decision Tree Algorithm.
13. What are Entropy and information gained in the Decision Tree Algorithm?
14. How do you manage an unbalanced data set?
15. What is lstm? How & Why was lstm used in your experience?
16. What are Recommender Systems?
17. What did you use to remove multicollinearity? Explain what values of VIF you used.
18. Explain different time series and analysis models. What are some time series models other than Arima?
19. How does a neural network with one layer and one input and output compare to logistic regression?
20. Describe Ensemble learning.
21. During analysis, how do you treat missing values?
22. How can outlier values be treated?

Is Machine Learning on your mind: Getting Started with Machine Learning Algorithms

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You can refer to these books to learn Machine Learning: [Top 25 Essential Machine Learning Books For Beginners & Experienced Professionals](#)

6] Amazon Data Scientist Interview Questions on Deep Learning:

1. What is Deep Learning?



2. What is the difference between Machine Learning and Deep Learning?
3. Why do you think Deep Learning is gaining more popularity?
4. Explain Neural Network fundamentals.
5. Describe Neural Networks.
6. Define reinforcement learning.
7. What are Hyperparameters?
8. Differentiate between Epoch, Batch, and Iteration in Deep Learning.
9. What is the structure of an Artificial Neural Network?
10. What are the different layers on CNN?
11. How does Pooling work on CNN?
12. Explain how the LSTM network works.
13. Define the Gradient Descent.
14. What are exploding gradients?
15. Explain Cost Function.

Master Deep Learning: [How to become proficient in deep learning and neural networks in just 30 days!!](#)

7] Amazon data scientist interview questions on Statistics:

1. What is the p-value?
2. What is the maximum likelihood of getting k heads when you tossed a coin n times?
3. There are 4 white balls and 2 green balls. What's the probability of them not being the same in the 2 picks?
4. How would you explain hypothesis testing for a newbie?



5. What is cross-validation?
6. How do you interpret OLS regression results?
7. Explain confidence intervals
8. Name the five assumptions of linear regression.
10. Estimate the disease probability in one city given the probability is very low nationwide. Randomly asked 1000 people in this city, with all negative responses (NO disease). What is the probability of disease in this city?
11. What is the difference between linear regression and a t-test?

Further here are some interview questions generally asked in Amazon Data Scientist interviews for checking your Leadership and Interpersonal skills.

Also find this: [What Skills Are Needed To Be A Data Scientist?](#)

8] Additional Amazon data scientist interview questions:

Tell me about yourself.

2. Where do you see yourself within the next 5 years?
3. Tell me a time when a goal was hard to achieve. What did you learn from that?
4. Why Data Science?
5. If you have to simplify some of your projects for customer experience enhancement, how would you proceed with it? Explain.
6. Tell me a time you used the data to come up with data-driven statistics, and how did you present your findings?
7. Describe the situation when you disagreed with your manager, and how did you handle that?
8. Share about any of your failed projects that might have been successful, with a few changes made to them.



9. Explain the most innovative idea you have ever had?
10. Tell me about a time you applied judgment to a decision when data was not available.
11. Tell me about your most important accomplishment. Why do you think it is important
12. What is the composition of your current team, and how are you encouraging their growth?
13. Describe the last time you figured out a way to keep an approach simple or to save on expenses
14. What is the one feedback/complaint you always get from your colleagues? How are you working on such feedback?

So, that makes a complete list of Amazon Data Scientist Interview Questions that you can master to crack the interview with not much pain.

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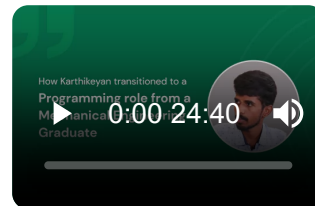
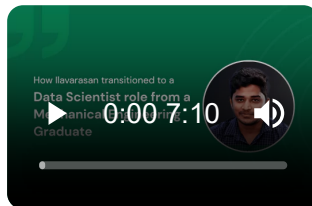
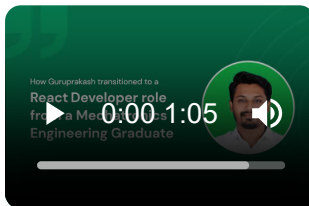




Don't forget to leave your queries and suggestions below in the comment section.

Further, if you require any more information regarding Amazon data scientist interview questions, [CHECK HERE](#).

Career transition



About the Author

Archana

A traveler, and explorer, Archana is an active writer at GUVI. You can usually find her with a book/kindle binge-reading (with an affinity to the mystery/humor).

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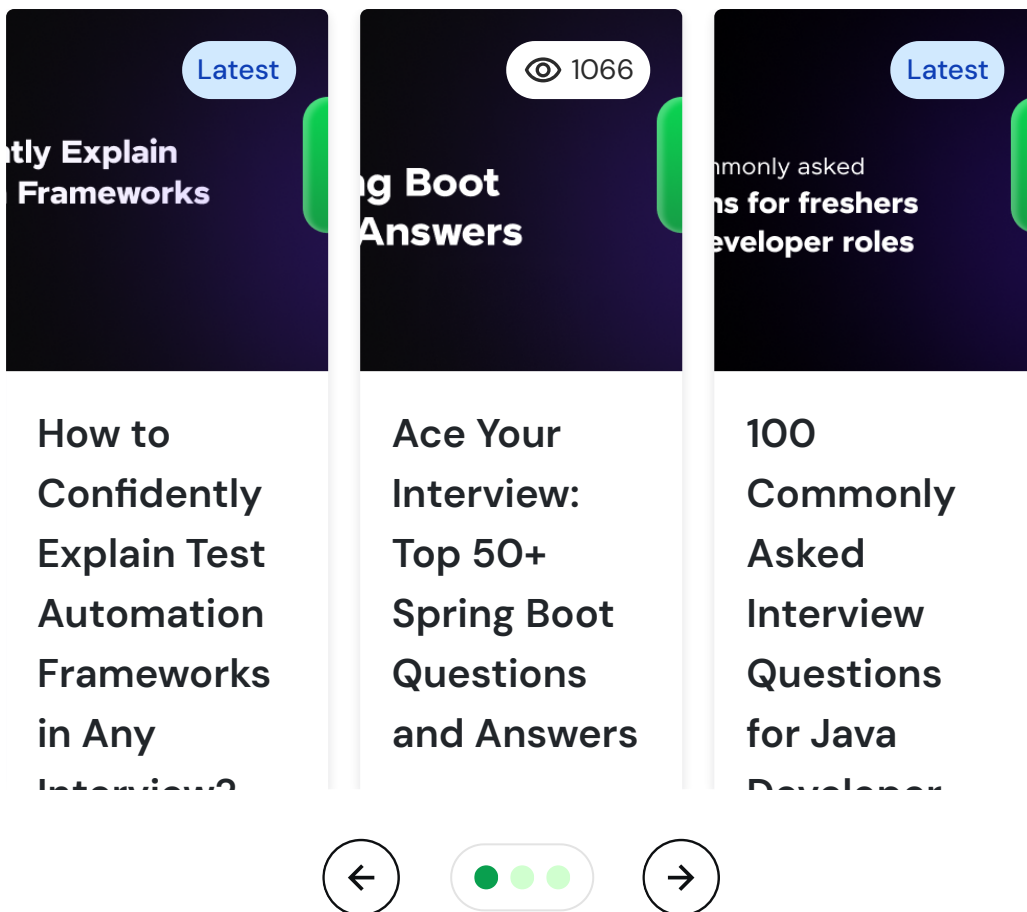


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